

Roll No.:

END SEMESTER EXAMINATION JUNE 2025

Name of the Program: B.Tech (CSE)

Semester: VI

Name of the Course: Software Engineering

Course Code: TCS 611

Time: 3 Hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	Define the term software engineering. Explain the common steps involved in developing a software.	C01
(b)	Discuss about the Software Crisis? Discuss in detail the problems and causes of software crisis. What is the significance of software crisis in reference to software engineering discipline.	C01
(c)	Explain the disadvantages of using waterfall model. Also explain the scenarios when we chose waterfall model for development.	C01
Q2		
(a)	Narrate the importance of software specification of requirements. Explain a typical SRS Structure and its parts.	C02
(b)	Name the responsibilities of the user, the customer and the developer in the requirements process?	C02
(c)	Explain about the Functional and non-functional requirements of software.	C02
Q3		
(a)	What is software Architecture style? Explain data centric/ driven architecture with an example.	C03
(b)	Explain data, architectural, interface and procedural design for a software?	C03
(c)	Explain modularity, coupling and cohesion and also establish relationship between these.	C04
Q4		
(a)	How can OOD be used to improve the maintainability and scalability of software?	C04
(b)	Discuss about the Top-down vs Bottom-up approach when designing a class library in software development.	C04
(c)	What is software metrics? Define function point and with the help of figure 1.0 calculate function point of a data flow model for safe home software.	C05

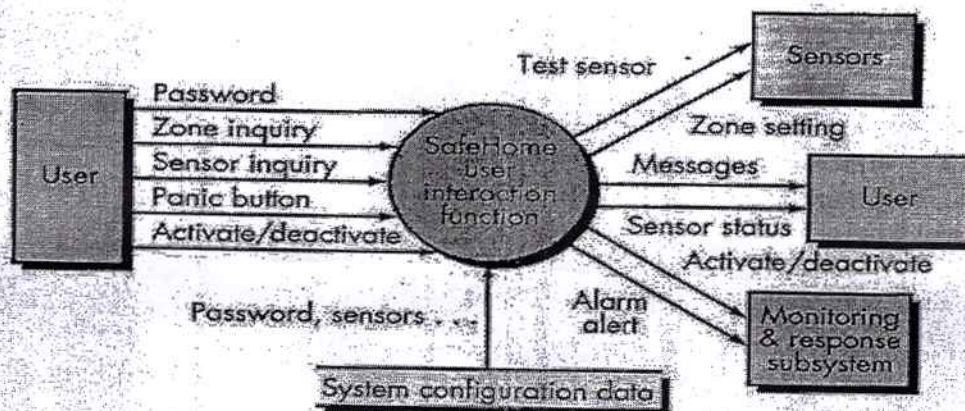


Figure 1.0 A Data Flow Model for Safe Home Software

Consider weighting factor of software is average, therefore, information domain values are as External interface is 4 external output is 5, External inquiries is 4, internal logical file is 10 and External interface file is 7. The Fi are value adjustment factors based on responses to the questions (suppose 18) are on average scaling factor.

Q5		
(a)	What are the levels of testing? Explain any two levels.	CO5
(b)	What is the role of test cases and test scenarios? Who should perform the validation test the software developer or software user justify your answer?	CO6
(c)	What are some cost drivers considered in Intermediate COCOMO? Discuss some advantages of using the COCOMO model	CO6